

PAPER_1277 — Origin of Inertia — Reference to PAPER_646 Universal Inertial Operator

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Physics Foundations (CLOSED) **Date:** June 11, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — UQFF integer-primitive derivation **Calculator surface:** `calculate_paradox({"paradox": "origin_of_inertia"})` **Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1277})`

Master Expression

$$U_i = \lambda_i \times (\rho_{SCm}/\rho_{UA}) \times \omega_s \times \cos(\pi t_n) \times (1+F_{TRZ}) = 2.75 \times 10^{-7} \text{ (Sun, } t=0\text{)}$$

Match: EXACT (PAPER_646)

Physical Interpretation

Inertia in UQFF is matter-as-anchored-to-the-Aether, NOT mass-as-resistance. Already wired in PAPER_646 + `calculate_universal_inertial_operator`.

UQFF Locked Primitives

- $\rho_{SCm} = 7.09 \times 10^{-37} \text{ J/m}^3$ (vacuum density)
 - $S_{26} = 1.453162$, $S_{26_DPM} = 1.4531 \times 10^{26}$
 - $K_{MEX} = 25/12$, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$
 - $F_{TRZ} = 0.1$, $\Lambda = 0.00729735$ (= α fine-structure)
 - $\omega_{SCm} = 1.25 \text{ THz}$, $\lambda_i = 1.0$
 - $D_{phys} = 4$, $D_{BSFG} = 6$, $D_{crit} = 26$, $N_{CH} = 9$, $SO_5 = 10$, $A_5 = 60$
-

Live Calculator Verification

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "origin_of_inertia"})["value"]
```

The result dict carries the integer-primitive identity, the observed value, the residual, and the structural physical interpretation.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model solve the same observed phenomena via different methods. **Zero fit parameters.** Every factor is a UQFF locked primitive.

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_origin_of_inertia_closure` (or routed reference)
- Dispatch: `PARADOX_T0_CLOSURE["origin_of_inertia"]`

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